

Class 10 Maths - Chapter Wise Detailed Notes

Real Numbers

Euclid Division Lemma: $a = bq + r$, $0 \leq r < b$.

HCF can be found using Euclid algorithm.

Irrational numbers cannot be written in p/q form.

Fundamental Theorem of Arithmetic: Every number has unique prime factorization.

Polynomials

Polynomial is expression of variables and constants.

Degree = highest power of variable.

Zeros of polynomial are values where polynomial becomes zero.

Relation: sum and product of zeros depend on coefficients.

Pair of Linear Equations

Two equations in two variables.

Methods: substitution, elimination, cross multiplication.

Solution is point where both lines intersect.

Quadratic Equations

Standard form: $ax^2 + bx + c = 0$.

Methods: factorization and quadratic formula.

Discriminant $D = b^2 - 4ac$ decides nature of roots.

Arithmetic Progression

Sequence with common difference.

nth term: $a_n = a + (n-1)d$.

Sum: $S_n = n/2 [2a + (n-1)d]$.

Triangles

Basic proportionality theorem.

Similarity of triangles rules.

Pythagoras theorem: $a^2 + b^2 = c^2$.

Statistics

Mean, median, mode.
Graphical representation of data.
Cumulative frequency.

Probability

Probability = favorable outcomes / total outcomes.
Value ranges from 0 to 1.
Simple events and examples.